Color Processor Chip

FEATURES

- Operation from 7.15909MHz crystal
- Five-line digital selection for 1 of 16 colors, blanking, Sync and color burst
- 3.579545MHz buffered output

DESCRIPTION

The required color to be displayed for each 280 nsec PIXEL is decoded on a four line binary coded input. This selects one of sixteen possible colors. An external resistor network completes the D to A function as shown in the schematic Fig. 1 of this document. The waveform plus table illustrates the use of the five inputs to produce composite sync, color burst, line blanking, frame blanking and video.

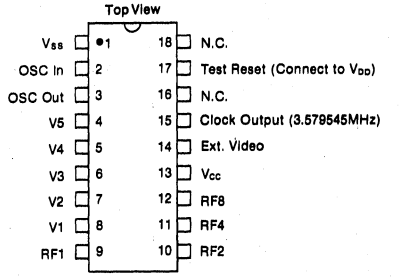
The external video input pin provides the ability to superimpose white high resolution (140 nsec wide) video information over the picture (color image).

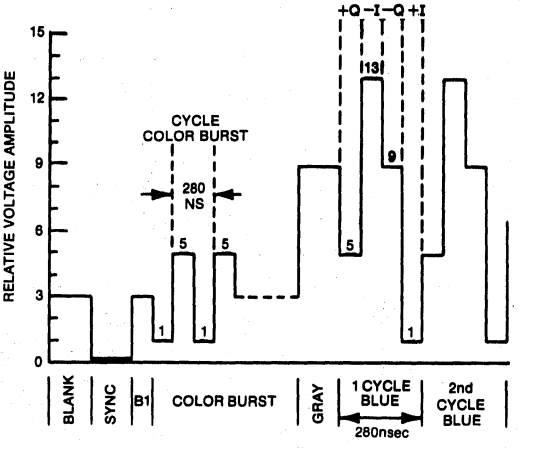
INPUT CODE ASSIGNMENT					TIME SLOT RELATIVE VOLTAGE AMPLITUDES				COLOR OUTPUT DESCRIPTION	
V5	V4	V3	V2	V1	+Q	−I	−Q	+I		
0	0	0	0	0	3	3	3	3	Black	
0	0	0	0	1	5	13	9	1	Blue	Group A
0	0	0	1	0	8	0	4	12	Red	
0	0	0	1	1	4	4	12	12	Tan	
0	0	1	0	0	3	8	11	6	Grass Green	
0	0	1	0	1	3	11	13	5	Green	
0	0	1	1	0	9	11	15	13	Yellow	
0	0	1	1	1	13	13	13	13	White	
0	1	0	0	0	9	9	9	9	Gray	
0	1	0	0	1	8	13	12	7	Cyan	Group B
0	1	0	1	0	9	4	9	14	Orange	
0	1	0	1	1	4	4	8	8	Brown	
0	1	1	0	0	13	5	3	11	Magenta	
0	1	1	0	1	12	12	6	6	Light Blue	
0	1	1	1	0	5	9	13	9	Yellow-Green	
0	1	1	1	1	10	5	2	7	Purple	
1	X	X	0	0	3	3	3	3	Blanking	
1	X	X	1	0	1	1	5	5	Color Burst	
1	X	X	0	1	0	0	0	0	Sync	
1	1	1	1	1	0	15	0	15	Test	

X = Don't Care

PIN CONFIGURATIONS

18 LEAD DUAL IN LINE





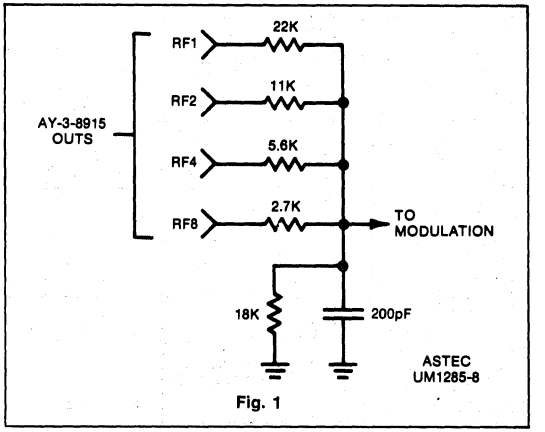


Fig. 1

ELECTRICAL CHARACTERISTICS

Maximum Ratings*

Temperature Under Bias ..... 0°C to +40°C
Storage Temperature ..... −55°C to +150°C
All Input or Output Voltages with Respect to V_{SS} ..... −0.2V to +9.0V
V_{CC} with respect to V_{SS} ..... −0.2V to +9.0V

* Exceeding these ratings could cause permanent damage. Functional operation of this device at these conditions is not implied—operating ranges are specified below.

Standard Conditions (unless otherwise stated)

T_A = 0°C to +40°C
V_{CC} = +4.85V − + 5.15V
V_{SS} = 0.0V

Characteristic	Sym	Min	Typ	Max	Units	Conditions
Oscillator Freq. In.	—	—	—	—	MHz	7.15909MHz crystal Trimmed by external capacitor
3.579545MHz Clock Output						
Output Logic Low	V_{OL}	0	—	0.5	volts	
Output Logic High	V_{OH}	2.4	—	V_{CC}	volts	
Logic Inputs V1, V2, V3, V4, V5, EXT. VIDEO						
Input Logic Low	V_{IL}	0	—	0.7	volts	
Input Logic High	V_{IH}	2.4	—	V_{CC}	volts	
Outputs RF1, RF2, RF4, RF8						
Output ON	I	5.0	—	—	mA	V_{out} = +0.5V
Output OFF	I	—	—	10	μA	V_{out} = +2.4V
Supply Current						
V_{DD}	I_{CC}	—	—	80	mA	V_{CC} = 5.25V @ 40°C